



CareConnect

Empowering Families to Manage Parent's Health — Smartly & Seamlessly

Field	Details
Document	Product Requirements Document (PRD)
Version	1.0
Date	March 31, 2026
Status	Active Development
Platform	Mobile App (iOS & Android)

1. Executive Summary

CareConnect is a mobile-first health management platform designed to help adult children and family members manage and monitor the health of aging parents — regardless of geographic distance. The app centralizes appointments, medical reports, medication reminders, and real-time health monitoring into one seamless, intelligent experience.

CareConnect removes the fragmentation of managing aging parent healthcare by bringing every touchpoint — doctors, reports, vitals, and family communication — into a single, AI-powered platform.

2. Problem Statement

Managing an aging parent's healthcare from a distance is overwhelming and fragmented. Adult children face four core pain points:

#	Pain Point	Description
1	Missed Appointments 	No central system to track doctor visits, follow-ups, or lab checkups
2	Lost Medical Reports 	Diagnosis reports pile up unorganized and are hard to interpret or share
3	No Reminders 	Medication schedules and test reminders are easily forgotten
4	No Real-Time Updates 	Family members living away have zero visibility into parent health status

3. Goals & Success Metrics

3.1 Goals

- Reduce missed medical appointments by 80%
- Enable remote family members to stay informed about parent health in real time
- Simplify medical report interpretation using AI and OCR
- Create a single source of truth for aging parent healthcare

3.2 Key Success Metrics

Metric	Target
Appointment booking activation (within 30 days)	70%+ of users
Reminder acknowledgement rate	> 85%
AI report analysis satisfaction rating	> 4.2 / 5
Family member daily active usage rate	> 50%
Report processing time	< 30 seconds
Notification delivery uptime	99.9%

4. Target Users

User Type	Role	Primary Interaction
Primary — Child/Guardian	Adult managing parents' healthcare, often remotely	Bookings, reports, dashboard, reminders

User Type	Role	Primary Interaction
Secondary — Parent/Patient	Elderly individual receiving care	Reminders, alerts, emergency SOS
Tertiary — Doctor/Specialist	Medical professional	Receives summaries, shares records

5. Features — Detailed Requirements

5.1 Feature 1: Appointment Booking

A complete doctor appointment management system enabling families to schedule, track, and manage all medical visits in one place.

Functional Requirements

- Search doctors by specialty, location, and availability
- Book, reschedule, or cancel appointments with one tap
- Auto-sync appointments to Google Calendar
- Upload prescriptions and medical history before the visit
- Receive digital consultation summaries post-visit
- Support both video and in-person appointment options

Booking Flow

Step	Action
1	Search Doctor / Clinic by specialty or location
2	Choose available Date & Time Slot
3	Add Patient Details (history, prescriptions)
4	Confirm & Pay (if applicable)
5	Receive Confirmation + Automatic Reminder

Non-Functional Requirements

- Appointment confirmation response time: < 3 seconds
- Google Calendar sync latency: < 5 seconds
- Doctor search results returned: < 2 seconds

5.2 Feature 2: AI-Powered Report Analysis

Users upload medical reports (photo or PDF) and receive plain-English AI summaries, flagging abnormal values and trends.

How It Works

Step	Process	Detail
1	Upload Report	Photo or PDF of lab/diagnostic report
2	OCR & Parse	Tesseract extracts values from the document
3	AI Analysis	Ollama model flags abnormal values, trends, risks
4	Plain English Summary	Easy-to-read results sent to all family members

Report Types Supported

- Blood Tests: CBC, Lipid Panel, Blood Sugar, Thyroid
- Radiology: X-ray, MRI, CT Scan summaries
- Trend Tracking: Values charted over time with visual graphs
- Abnormality Alerts: Instant push + SMS notification for critical values

Non-Functional Requirements

- Report processing time: < 30 seconds end-to-end
- OCR accuracy: > 92%
- AI summary must include a confidence score

5.3 Feature 3: Smart Reminders & Alerts 🕒

A configurable, multi-channel reminder engine covering all health-related tasks for both the parent and connected family members.

Reminder Types

Type	Schedule	Icon
Medication Reminders	Daily, weekly, or fully custom schedule	
Doctor Follow-ups	Automatic reminder post-appointment	
Lab Test Reminders	Includes fasting and prep instructions	
Vaccination Alerts	Flu shots, boosters, and more	

Smart Alert Capabilities

- Push notifications and SMS alerts via Twilio
- Multi-family member simultaneous notifications
- Snooze and reschedule with one tap
- Doctor-recommended frequency auto-configured

- Missed reminder escalation alerts
- Voice reminders for elderly parents

Non-Functional Requirements

- Notification delivery latency: < 10 seconds
- System uptime for reminders: 99.9%
- Support 10+ concurrent reminder schedules per patient

5.4 Feature 4: Health Monitoring Dashboard

A visual health dashboard aggregating all vitals, reports, and appointment history in one place with trending charts and doctor-share functionality.

Vitals Tracked

Vital	Unit	Normal Range
Heart Rate 	bpm	60 – 100 bpm
Blood Pressure 	mmHg	< 120/80 mmHg
Blood Sugar 	mg/dL	70 – 140 mg/dL (post-meal)
BMI 	kg/m ²	18.5 – 24.9

Dashboard Capabilities

- Vitals Trending: Visual graphs of BP, sugar, and weight over weeks/months
- Report Repository: All diagnostic reports in one searchable, organized timeline
- Visit History: Complete log of all doctor appointments and outcomes
- Share with Doctor: One-click health summary export (PDF/CSV) for consultations

Non-Functional Requirements

- Dashboard load time: < 2 seconds
- Chart rendering time: < 500ms
- Export available in PDF and CSV formats

5.5 Feature 5: Family Sharing

Multi-user access enabling all family members to stay updated on the parent's health status in real time.

- Invite family members via email or shareable link
- Role-based access: Admin (full control) and Viewer (read-only)
- Real-time push notifications to all connected family members

- Activity log showing who viewed or modified patient data

5.6 Feature 6: Emergency Alerts 🚨

Instant SOS alerting system that notifies all family members and emergency contacts during a health emergency.

- One-tap SOS button on the parent's app interface
- Automatic alert escalation if no family member responds within 5 minutes
- Live location sharing during an emergency
- Pre-configured emergency contacts list
- Integration with local emergency services directory

6. Technology Stack

Layer	Component	Technology
Frontend (Mobile)	Framework	React Native with Expo
Frontend (Mobile)	State Management	Redux Toolkit
Frontend (Mobile)	UI Library	React Native Paper
Backend API	Runtime	Python with Flask
Backend API	API Style	REST API
Backend API	Authentication	JWT (JSON Web Tokens)
AI & ML Layer	LLM Inference	Ollama (local)
AI & ML Layer	OCR Engine	Tesseract OCR
AI & ML Layer	NLP	Custom NLP Models
Database & Storage	Primary Database	PostgreSQL
Database & Storage	File Storage	Firebase Storage
Database & Storage	Cache	Redis
Integrations	Calendar Sync	Google Calendar API
Integrations	SMS / Voice	Twilio
Integrations	Doctor Data	Hospital APIs

7. Development Roadmap

Phase	Focus Area	Deliverables
Phase 1Foundation	Core Setup	• Requirement gathering & wireframes• UI/UX design system• Core auth & user profiles• Appointment booking MVP
Phase 2Core Features	AI & Reminders	• Report upload & OCR pipeline• AI-powered report summaries• Reminder system (push + SMS)• Family multi-user access
Phase 3Advanced	Dashboard & Video	• Health dashboard & vitals tracking• Calendar sync integrations• Video consultation module• Beta testing & feedback
Phase 4Launch	Production	• Performance optimization• Hospital API integrations• Emergency alert system• App Store & Play Store launch

8. Security & Compliance

- All data encrypted at rest (AES-256) and in transit (TLS 1.3)
- HIPAA-compliant data handling practices
- JWT tokens with 24-hour expiry and refresh token rotation
- Role-based access control (RBAC) enforced at API layer
- Full audit logs for all data access and modification events
- Biometric authentication support (Face ID / Fingerprint)

9. Out of Scope (v1.0)

- Insurance claim processing
- Telemedicine billing integration
- Wearable device sync (planned for v2.0)
- Multi-language support (planned for v2.0)
- AI-generated treatment recommendations

10. Launch Targets

Milestone	Target
Finalize wireframes and design system	Week 1–2
Set up development environment & tech stack	Week 2–3
Build MVP — Appointment Booking + Reminders	Week 4–7
Integrate AI Report Analysis module	Week 8–10
Beta launch with 50 target families	Week 12
App Store & Google Play Store release	Week 16

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